Designing a DEER-Toolkit: Enabling Educational Escape Rooms in a 2D Dungeon Crawler Game

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Abstract

Digital Educational Escape Rooms (DEERs) have gained increasing attention as an engaging form of game-based learning. However, existing solutions often rely on web-based point-and-click interfaces that offer limited immersion, or on virtual reality implementations that require substantial development effort and specialized hardware. This work introduces a DEER-Toolkit integrated into a self-developed 2D multiplayer dungeon crawler game environment, designed to balance accessibility, interactivity, and educational value. The toolkit provides configurable puzzle mechanics, collaborative gameplay elements, and lightweight learning analytics to support facilitation and post-play reflection. To illustrate its capabilities, a short cybersecurity-themed DEER scenario is presented, designed for one to three players and playable within ten minutes. The demonstrator highlights how narrative framing, interactive object-based puzzles, and adaptive scaffolding can support active learning in a compact format suitable for classroom or workshop use. Future work focuses on developing authoring tools to enable educators to create their own scenarios and on evaluating the toolkit's effectiveness in authentic teaching contexts.

CCS Concepts

• **Applied computing** → *Collaborative learning*; *E-learning*; **Interactive learning environments**.

Keywords

Digital Educational Escape Room (DEER), Game-based learning, Narrative-centered learning,

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1 Introduction

Traditional instructional formats in higher education, such as lectures and exercise classes, increasingly fall short of engaging contemporary learners and fostering sustained interaction with disciplinary content [12]. Today's student cohorts expect active, participatory, and multimedia-enhanced learning experiences that emphasize personal relevance, collaboration, and achievement [1, 11]. As a result, interest in game-based approaches that enrich learning environments while maintaining pedagogical rigor has intensified.

Game-Based Learning (GBL) is a pedagogical approach that examines how the motivational and interactive properties of games can be leveraged to support knowledge acquisition and skill development. Rather than serving merely as a means of assessment, GBL integrates learning objectives directly into the structure of a game or playful activity in order to foster engagement, motivation, and meaningful understanding [10]. Digital Educational Escape Rooms (DEERs) form a specific category of GBL. They draw on the concept of recreational escape rooms, in which teams solve interconnected puzzles within a limited timeframe to achieve a shared objective, and integrate explicit learning outcomes directly into this structure [17]. Empirical studies show that DEERs can increase learner motivation, foster collaboration, and promote deeper cognitive engagement compared to conventional learning activities [16, 29].

Prior research on DEERs has primarily focused on conceptual and pedagogical frameworks that guide educators in the design and planning of escape-room-based learning activities. Prominent examples include EscapED [4] and Room2Educ8 [7], which provide

structured, learner-centred methodologies for defining learning objectives, designing puzzles, and embedding escape rooms within educational contexts. These frameworks make an important contribution by clarifying how DEERs can be pedagogically grounded and systematically designed. However, they deliberately abstract from the technical execution layer and do not provide concrete, reusable tools for implementing DEERs as interactive game systems with built-in support for collaboration, adaptivity, or learning analytics.

In recent years, web-based DEER platforms have made the creation of escape-room activities more accessible for educators who lack programming expertise [30]. Popular tools include Genially [20], BreakoutEDU [21], and Google Forms [2, 3]. These platforms provide browser-based puzzle design through intuitive interfaces and integrated automation features such as progress control and basic assessment [2, 13]. However, they typically depend on cloud deployment, which limits data privacy and restricts local use, and their audiovisual capabilities constrain immersion. Point-and-click mechanics offer limited interactivity and environmental complexity. As a result, it is difficult to create convincing game worlds that evoke a strong sense of presence, which has been identified as an important factor in the effectiveness of DEERs [25, 29].

Immersive VR-based DEERs provide more embodied interaction and rich sensory environments [16]. Despite these advantages, VR implementations introduce several barriers. They require considerable development effort and specialized hardware, which reduces accessibility and scalability. Novice users often face usability challenges, including motion sickness and difficulties navigating virtual environments [5, 24, 27]. Furthermore, VR platforms generally offer fewer authoring options and less integrated support for automated evaluation than web-based solutions, which limits their practicality for many educational settings.

In this context, a gap remains between high-level pedagogical DEER frameworks and concrete technical solutions that enable DEERs to be executed as reusable, interactive, and data-driven game systems. This paper addresses this gap by presenting the conceptual design and planned functionality of a DEER-Toolkit embedded in a self-developed 2D dungeon crawler multiplayer game environment, a game genre characterized by the cooperative exploration of interconnected rooms, object-based interaction, and puzzle-driven progression. The proposed toolkit aims to combine the accessibility and low hardware requirements of 2D games with a higher degree of interactivity and immersion than conventional point-and-click DEERs, while avoiding the accessibility barriers associated with VR-based approaches. It provides configurable gameplay elements for implementing educational puzzle mechanics across disciplines, supports synchronous collaboration among multiple participants, and incorporates adaptive hinting and integrated learning analytics to capture and automatically evaluate player actions and team dynamics. The overarching research goal is to explore how pedagogical DEER concepts can be operationalized within a reusable, analytics-enabled game-based execution environment that complements existing design-oriented frameworks.

The contribution of this project is twofold. First, it introduces a custom-built, non-VR DEER environment that extends beyond web-based platforms by enabling dynamic interaction, spatial navigation, and richer audiovisual storytelling. Second, it outlines a toolkit that supports instructors in creating domain-specific DEER scenarios without requiring extensive programming expertise. The demonstrator will present the core game environment, puzzle mechanics, collaborative features, and analytics components, illustrating how the platform can be used to design engaging educational escape-room experiences.

This work focuses on the conceptual foundations and system design of the DEER-Toolkit. Authoring tools for scenario creation are part of the broader project but are not addressed here and will be presented in future work.

2 Concept

The *Dungeon framework*[1] is a Java-based framework that provides the technical and gameplay infrastructure for a 2D dungeon crawler environment. Its open-source nature enables continuous refinement, community contributions, and long-term maintainability. The framework is lightweight and has low hardware requirements, supporting deployment in typical educational settings such as computer labs and classrooms. As it is developed and actively maintained by the authors, DEER-specific features can be integrated directly into the framework and iteratively refined based on evaluation outcomes. These characteristics make the Dungeon framework a suitable foundation for the development of DEERs, offering a flexible and familiar game structure that can be extended with educational puzzles, multiplayer collaboration, and integrated learning analytics.

The DEER-Toolkit constitutes a set of dedicated functionalities, components, and content modules within the Dungeon framework that enable the design and implementation of DEERs. DEER scenarios can be deployed in both instructor-led sessions and self-directed learning formats. While the primary target context is higher education, the DEER-Toolkit does not impose domain restrictions and can also be applied in other educational or training settings.

2.1 Definition and Educational Positioning of Escape Rooms

Nicholson (2016) [18] defines escape rooms as live-action, team-based games in which players uncover clues, solve puzzles, and complete tasks across one or more rooms in order to achieve a specific goal within a limited time. In line with this definition, this work conceptualizes DEERs as collaborative, time-bounded problem-solving activities embedded within an interactive game world, in which puzzle completion and coordinated actions determine progression.

From an educational perspective, DEERs are particularly well suited for the consolidation and application of previously introduced content, rather than for the initial acquisition of new knowledge.

A systematic review indicates that escape rooms are effective in enabling learners to experience and practice concepts, although evidence for substantial knowledge gains measured through pre- and post-assessments remains limited. At the same time, engagement in escape rooms correlates positively with self-perceived learning effectiveness [30]. These findings support a design approach that

[1]The Dungeon framework is available as open-source software on GitHub: https://github.com/Dungeon-CampusMinden/Dungeon.

prioritizes active engagement, concept reinforcement, and structured debriefing to facilitate knowledge transfer.

2.2 Puzzle Taxonomy and Required Competencies

Puzzle design in the DEER-Toolkit is guided by commonly observed puzzle categories in escape rooms [18]. The framework therefore provides building blocks for:

- Search and discovery tasks involving objects or information within the environment.
- Logic- and pattern-based puzzles, such as symbol matching, sequence deduction, or basic mathematical reasoning.
- Hand-eye coordination challenges that require timing or spatial manipulation.
- Orientation and navigation through maze-like environments.
- Communication-dependent puzzles that require information sharing across roles or locations.

These puzzle types foster teamwork, communication, delegation, as well as critical and lateral thinking and attention to detail [18]. The DEER-Toolkit is designed to support combinations of multiple puzzle types in order to encourage collaboration and distributed problem solving.

2.3 Core Game Mechanics and Design Rationale

The game mechanics of the Dungeon framework and the DEER-Toolkit are informed by evidence-based recommendations aimed at supporting educational effectiveness and practical classroom implementation.

Hint System. A gradual and adaptive hint strategy is employed to maintain learner autonomy while preventing prolonged stagnation. Hints are released progressively to guide players toward the intended solution path without revealing it entirely. Adaptivity is informed by performance indicators derived from in-game behavior. Hint states are modeled using Petri nets, which enable explicit control of prerequisite conditions and ensure consistency across parallel puzzle tracks [26]. This approach supports differentiated scaffolding and reduces the need for instructor intervention during gameplay.

Time Management. A visible timer structures the gameplay experience and focuses attention, which can enhance immersion and goal orientation [29]. Scenarios are designed for completion within one to two hours in order to balance depth with the risk of fatigue, following guidance suggesting that shorter durations may limit meaningful learning, whereas considerably longer sessions may reduce engagement [8, 19]. This time allocation also reserves space for post-game reflection, which is necessary for linking in-game actions to learning outcomes [19].

Since DEERs are particularly effective for practice and consolidation, scenarios will align puzzle mechanics with previously taught concepts. A structured debriefing is essential, as time pressure during play can reduce opportunities for explanation and reflection [29]. The built-in timer ensures that time remains available for post-game discussion, during which instructors can connect gameplay experiences to learning objectives and address misconceptions.

Multiplayer and Group Size. The framework supports cooperative play with flexible team sizes. Configurations with two players allow both participants to engage with all puzzles [15]. Groups of three to six players are commonly used in educational escape rooms and are supported by the framework, as they can foster participation and immersion [30]. Larger groups are possible but may lead to reduced communication and coordination [6]. These group-size considerations are reflected in the design of the scenario templates and analytics parameters within the framework.

Points and Feedback. The scoring subsystem is optional and adaptable. While some learners are motivated by point systems, others may find them discouraging [19]. For this reason, the DEER-Toolkit supports both mastery-oriented feedback and progress-based reward elements. Prior work indicates that carefully designed scoring systems can positively influence learning outcomes and engagement [13]. The DEER-Toolkit provides live dashboards that allow instructors to monitor team progress, enabling targeted facilitation and timely support for struggling groups [14].

Interaction Primitives. The DEER-Toolkit includes standardized interaction types such as item crafting and combination, code and symbol locks, switches, and decorative or distractor objects to support environmental storytelling. These primitives allow puzzle authors to design consistent and intuitive interactions, thereby keeping the cognitive challenge focused on reasoning and collaboration rather than on interface complexity.

Player Roles and Personalization. The DEER-Toolkit offers optional elements that can be used to address diverse learner motivations, drawing on concepts from the Hexad player type model [28]. Prior work indicates that aligning game mechanics with user preferences can enhance engagement [21, 22]. Accordingly, educators and designers may include mastery-oriented challenges for Achiever-type learners, integrate collaboration-focused tasks for Socialiser preferences, provide team-support actions aligned with Philanthropist motivations, add exploratory elements for Free Spirits, incorporate reward-based progression for Player types, or offer unconventional challenge routes to engage Disruptor tendencies. These optional elements can be combined or adapted as needed. Any role-related features are intended to encourage complementary collaboration rather than competitive dominance.

Audio and Immersion. Sound design contributes to perceived quality and immersion in interactive environments [29]. The DEER-Toolkit integrates thematic soundscapes and contextual audio cues to enhance atmosphere and support environmental storytelling [9]. Spatial or directional audio can be employed to support certain puzzle types; however, such features are considered optional enhancements rather than core mechanics [31].

Communication Tools. The need for integrated voice chat depends on the usage context. In on-site settings, co-located communication may be sufficient, whereas integrated voice features can support remote or hybrid implementations. This functionality is evaluated during classroom deployments.

Learning Analytics. Web-based DEER platforms commonly provide event logging and dashboards that allow instructors to monitor progress and support learners during gameplay [13, 14]. In line

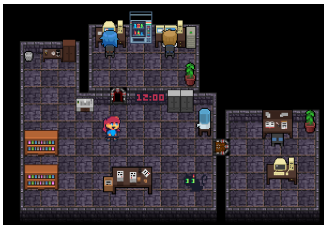

Figure 1: In-game screenshot of the *"The Last Hour"* demonstrator scenario. Players explore a research facility and interact with terminals, documents, and device interfaces to uncover clues and restore system access.

with this practice, the DEER-Toolkit records interaction data in a Learning Record Store using the xAPI standard to ensure interoperability with existing analytics systems [21, 23]. The aim is to provide lightweight, real-time visualizations of team activity so that instructors can identify potential difficulties and intervene when appropriate. Beyond these core features, additional indicators related to engagement or frustration are included only if they can be derived transparently from process data and applied meaningfully in instructional contexts [21].

3 Demonstrator

To illustrate the capabilities of the DEER-Toolkit, this work presents a short DEER titled *"The Last Hour"*[2]. The demonstrator is designed as a 10-minute guided experience for one to three players. It provides a compact example of how the toolkit combines narrative design, interactive puzzle mechanics, collaboration, and learning analytics within a dungeon-style game environment.

The scenario is set in an abandoned underground research facility where a scientist working on a confidential project has gone missing under unclear circumstances. Players enter the facility to stabilize its compromised systems and recover the scientist's classified research. The experience aims to reinforce essential cybersecurity awareness skills through applied problem-solving. Players are encouraged to:

- identify and evaluate information from multiple sources,
- distinguish legitimate from manipulated digital communication,
- apply safe data handling practices (e.g., password protection and disposal of sensitive information),
- reason collaboratively under time pressure, and
- make use of available hints and recovery strategies when needed.

Figure 1 shows the core game interface. Players explore an environment that contains interactive objects relevant to the narrative scenario. These include terminals that require secure authentication, documents and notes providing partial clues, locked storage elements, and digital interfaces such as simulated communication systems or data files. Together, these elements form the basis for interconnected puzzles that must be solved in order to progress.

[2]The Last Hour is available as open-source software on GitHub: https://github.com/Dungeon-CampusMinden/Dungeon/tree/master/theLastHourEscapeRoom.

The scenario highlights three core design elements enabled by the DEER-Toolkit:

(1) **Narrative Integration:** All puzzles are embedded in a coherent story that motivates player actions and provides context for information handling and decision-making.
(2) **Authentic Interaction Patterns:** Players engage with representations of real-world cybersecurity practices, such as reconstructing secure passwords from fragmented information, evaluating message authenticity, and handling removable storage devices safely.
(3) **Adaptive Support:** Optional scaffolding elements, such as progressive hints and recovery actions, allow players to continue after incorrect decisions or delays without interrupting the flow of the experience.

The demonstrator can be experienced individually or in small groups, supported by a brief facilitator introduction and a short post-play reflection. It illustrates:

- how narrative and puzzle mechanics are implemented using the toolkit,
- how the environment supports short, low-threshold DEER activities, and
- how lightweight analytics data is captured during play.

Following each play session, selected logged events (e.g., time spent, hint use, and error states) can be displayed to illustrate how instructors may analyze player behavior and identify areas for targeted feedback or debriefing. The demonstrator serves as an initial reference implementation that can be adapted to other domains and difficulty levels, demonstrating how the DEER-Toolkit can be used to create short, domain-relevant DEER experiences that combine narrative framing, interactive puzzles, and embedded learning support.

4 Future Work

As this project is currently in an early development stage, several technical and pedagogical aspects require further investigation. A primary direction for future work concerns the integration of authoring tools that enable educators to design DEER scenarios without programming expertise. While the current demonstrator showcases the gameplay, interaction mechanics, and analytics infrastructure of the DEER-Toolkit, the creation of new scenarios still requires technical expertise. Developing an intuitive, low-threshold editor is therefore essential to support broader adoption and to enable instructors from various disciplines to design domain-specific experiences.

A second focus lies in the systematic evaluation of the toolkit in authentic teaching contexts. Upcoming studies will examine how effectively the DEER-Toolkit supports learning processes, collaboration, and engagement, as well as how instructors make use of the available analytics during facilitation and debriefing. Of particular interest is how adaptive scaffolding and recovery mechanisms influence the balance between productive struggle and cognitive overload. Classroom pilots across different subject areas will inform refinements to puzzle design guidelines, hint logic, and analytics visualizations.

Overall, future development will focus on enabling scalable scenario authoring, validating educational impact through empirical

studies, and enhancing flexibility across different instructional settings. Expanding the library of reusable puzzle components and narrative templates is expected to support educators in creating varied and scalable escape room experiences. Together, these steps aim to transform the DEER-Toolkit from a research prototype into a robust resource that supports educators in implementing engaging, data-informed DEERs.

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